

ZERO ENERGY — PROTOCOL TECHNICAL SPECIFICATION

Version 1.0

PRELIMINARY NOTE

This document is the functional specification of the ZERO Energy protocol and the single reference source for three audiences: the community that participates in the ecosystem, the team that develops the protocol, and the auditors who verify it. It describes how the protocol works through rules, flows and equations.

Three reading rules:

- Where the document describes a limit, that limit is a mandatory rule of the protocol, not an intention.
- The **invariants** cannot be modified by any mechanism. The **parameters** may be configured by governance within the indicated ranges. Section 11 gathers them in two tables: invariants (11.1) and parameters (11.2).
- The evolution of the protocol is decided by the DAO through voting, as the ecosystem grows.

Nomenclature. The assets are identified as **ZERO DAO (ZDAO)** and **ZERO REC (ZREC)**. After the first mention in each section, the tickers are used. Formulas, tables, diagrams and contract specifications always use ZDAO and ZREC.

TOKEN ZERO is the Presale denomination. A TOKEN ZERO is a **future minting right**: it is neither a minted token nor a third on-chain asset. That right is exercised at the Claim, where it is resolved into ZDAO or ZREC at the participant's choice (Section 5).

Reference version. This English version is the reference version of the specification, published in the protocol's official repository. The Spanish and Portuguese versions are translations provided for ease of reading; in case of discrepancy, this English version prevails.

Extensions. Anything incorporated into the protocol after this version will be added as a separate, dated annex, without modifying the body of the document.

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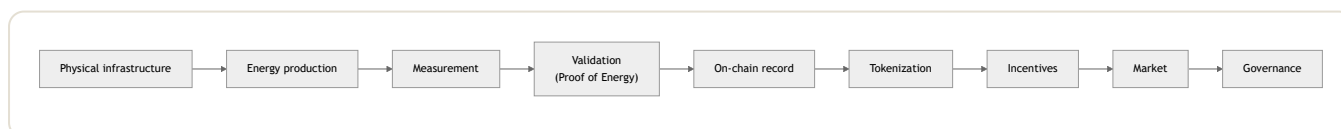
Note. This index refers to the files of the official ZERO Energy repository on GitHub, where each section of the document is published separately, in English (reference version), together with the complete document in Spanish, English and Portuguese: [https://github.com/ZEROEnergyDAO/ZERO-Energy-Docs/tree/main/04 Protocol](https://github.com/ZEROEnergyDAO/ZERO-Energy-Docs/tree/main/04%20Protocol)

1. GENERAL ARCHITECTURE

1.1 What ZERO Energy is

ZERO Energy is a **decentralized physical infrastructure network (DePIN) applied to the energy sector**, deployed on the Base network.

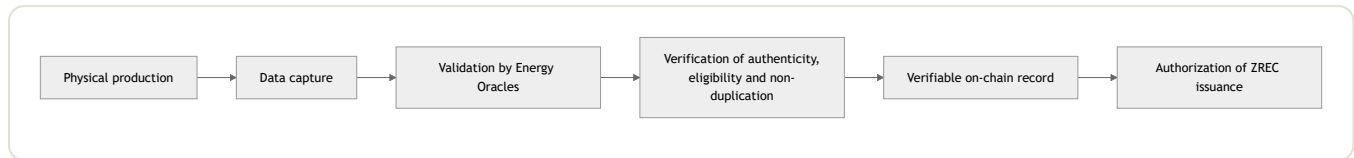
The protocol connects real renewable generation installations with a verifiable digital economy. The value chain is:



DePIN is neither an add-on component nor a commercial label: it is the architectural framework within which everything else operates. Where more precise, ZERO Energy may be described as an **energy DePIN**.

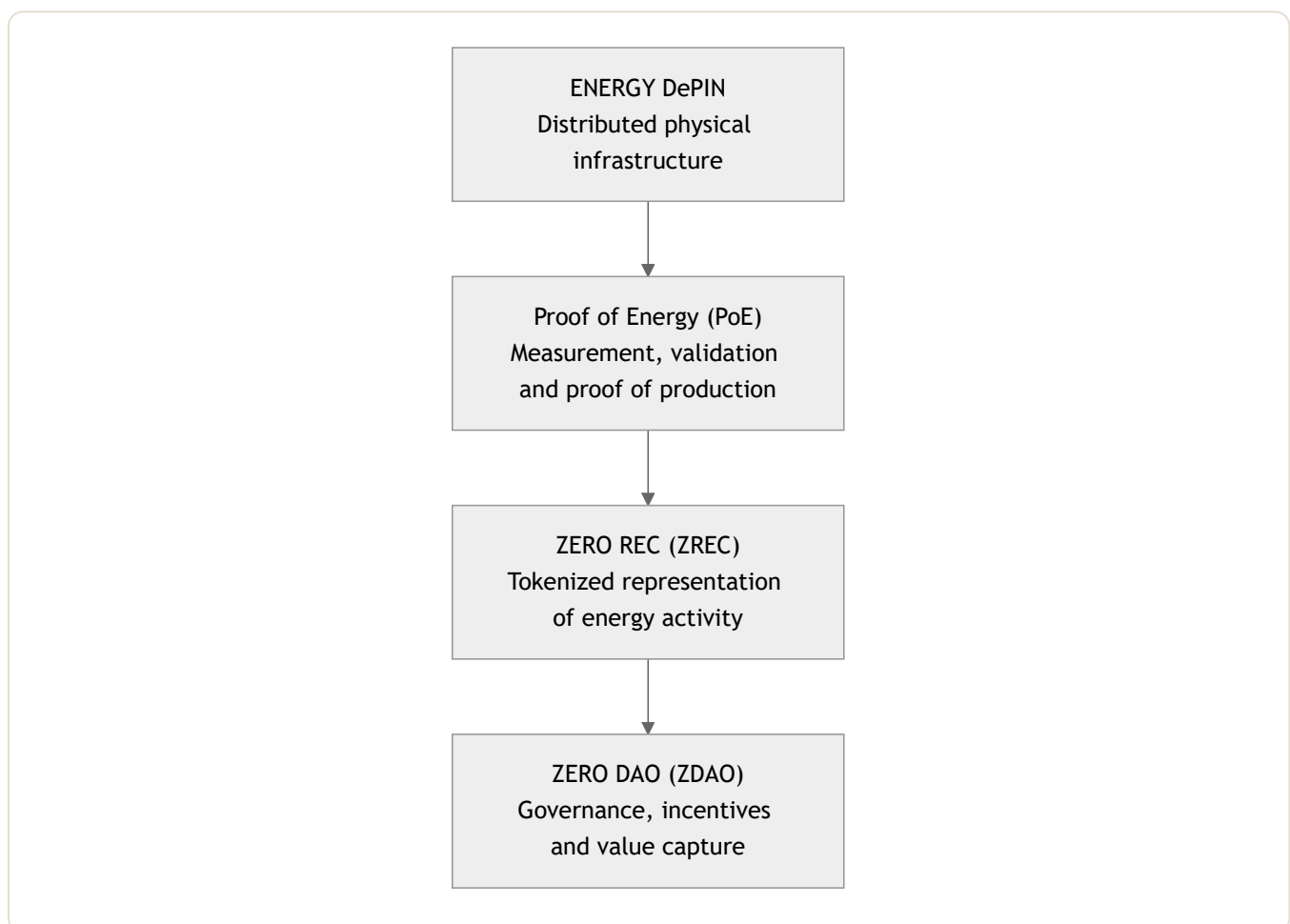
1.2 Proof of Energy (PoE)

Proof of Energy is the verification model through which physical energy production becomes a verifiable on-chain activity. It is the layer that connects the physical infrastructure with the protocol's digital economy.



Mandatory delimitation. Proof of Energy **is not a blockchain consensus mechanism**. Base maintains its own consensus and its own security. PoE operates in the protocol's energy validation layer and enables on-chain actions; it neither replaces nor modifies the network's consensus. This distinction must be maintained throughout all of the protocol's technical communication.

1.3 The two assets



	ZERO DAO (ZDAO)	ZERO REC (ZREC)
Function	Governance and incentives	Energy and utility asset

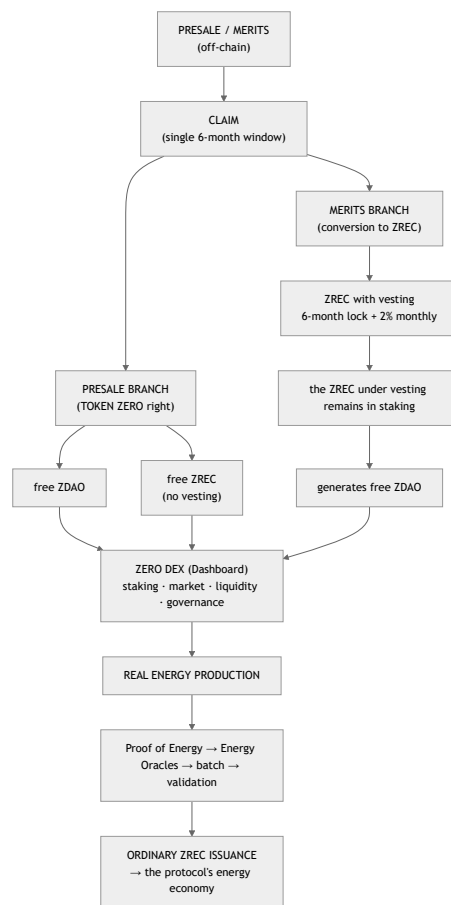
	ZERO DAO (ZDAO)	ZERO REC (ZREC)
Supply	Absolute maximum of 1,000,000,000	No predefined maximum
Decimals	8	3
Origin	Presale (TOKEN ZERO), Merits, Staking, Strategic Issuances	Presale (TOKEN ZERO), Merits and certified production
In staking	Received as a reward	The asset that is deposited

Structural principle. ZDAO maintains a strict issuance limit; ZREC expands as certified energy production grows. This separation allows the energy infrastructure to grow without compromising the scarcity of the governance asset. It is the economic foundation of the protocol and must not be altered.

1.4 Protocol layers

Layer	Component	Section
Physical	Installations, meters, inverters	4
Validation	Energy Oracles, Proof of Energy	4
Assets	ZDAO and ZREC	2 and 3
Onboarding	Presale, TGE and Claim	5
Incentives	Official Staking System	6
Interface	ZERO DEX (Dashboard)	7
Governance	DAO, Committee, decentralization milestones	8

1.5 Complete operating cycle



Each element of this cycle is developed only once in the document, in its corresponding section.

2. ZERO DAO (ZDAO)

ERC-20 governance and incentive token of the protocol, deployed on Base. **8 decimals**: the minimum unit is 0.00000001 ZDAO, the granularity required for the proportional calculation of rewards, fees and distributions.

2.1 Supply magnitudes

The document distinguishes the following magnitudes, which must not be confused:

Magnitude	Definition
MAX_SUPPLY	1,000,000,000 ZDAO. Absolute ceiling; it lives in the ZDAO ERC-20 token. It can never be exceeded
totalSupply	ZDAO in existence at this moment (minted minus burned). It is the token's standard circulating metric
Circulating supply	Equal to totalSupply: ZDAO has no locked positions

Magnitude	Definition
Minting capacity	<code>MAX_SUPPLY - totalSupply</code> , calculated in the ERC-20 itself. It is recovered when ZDAO are burned
<code>stakingMinted</code>	Total ZDAO minted by the staking contract. It only grows: burning does not reduce it. It lives in the staking contract, capped at 500M (Section 6.3)
Off-chain allocations	Minting rights registered in the Presale (denominated TOKEN ZERO) and in Merits, not yet minted
Burned ZDAO	Permanently destroyed; they reduce <code>totalSupply</code>

Invariants:

- **ERC-20 token:** `totalSupply ≤ MAX_SUPPLY`, checked on every mint.
- **Staking contract:** `stakingMinted ≤ 500,000,000`, checked on every reward mint.

Separation of responsibilities. `MAX_SUPPLY` lives in the ERC-20 token and bounds the global supply; `stakingMinted` lives in the staking contract and bounds only the issuance of rewards. ZDAO **has no locked positions**: staking rewards are free and immediately transferable (Section 6.5), and the Merits lock applies to the deposited ZREC principal, never to ZDAO.

ZDAO is **mintable and burnable** according to the standard ERC-20 mintable/burnable capability: any holder may burn their own ZDAO, and burning reduces `totalSupply`, which returns minting capacity under `MAX_SUPPLY`. The staking reserve (500M) and the decreasing issuance curve (Section 6.3) bound the issuance of rewards and are accounted for with `stakingMinted`, which burning does not roll back.

2.2 Burning returns minting capacity to the token, but not to the staking cap

Burned ZDAO are permanently destroyed and **reduce `totalSupply`**, thereby **returning minting capacity** under the ERC-20 `MAX_SUPPLY`.

Example:

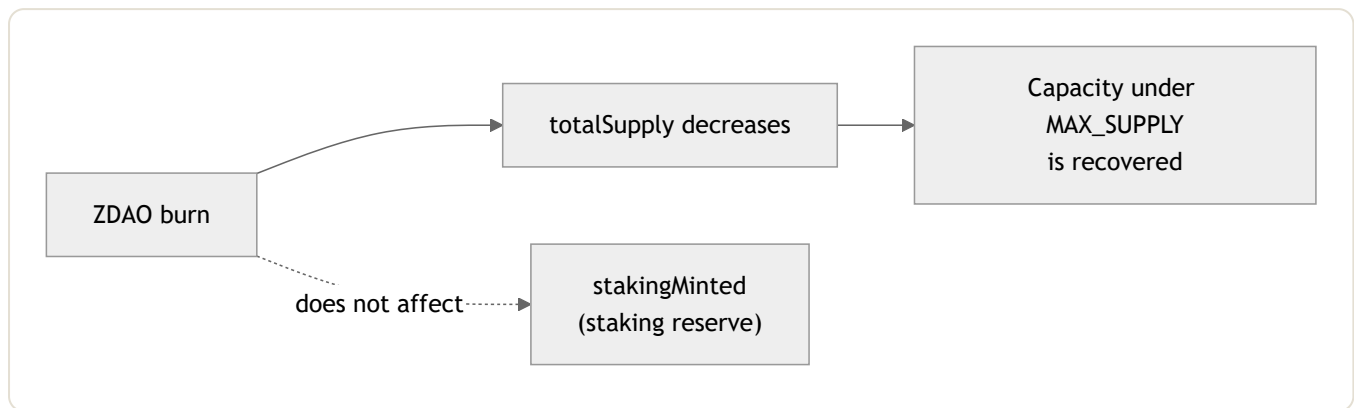
```
MAX_SUPPLY ..... 1,000M
totalSupply ..... 600M
Burned ..... 100M
totalSupply (after) ..... 500M
```

Remaining minting capacity = $1,000M - 500M = 500M$
(the 100M burned return as available capacity)

This is the standard ERC-20 mintable/burnable pattern: capacity is measured against the circulating `totalSupply`, so burning frees it.

The staking cap is different. `stakingMinted` is a monotonic counter of the staking contract (capped at 500M, Section 6.3). Burning ZDAO—including ZDAO that come from staking rewards— does **not** reduce

stakingMinted, and therefore does **not** replenish the staking reserve or the decreasing curve. The staking contract can never mint more than 500M in total, however much is burned.



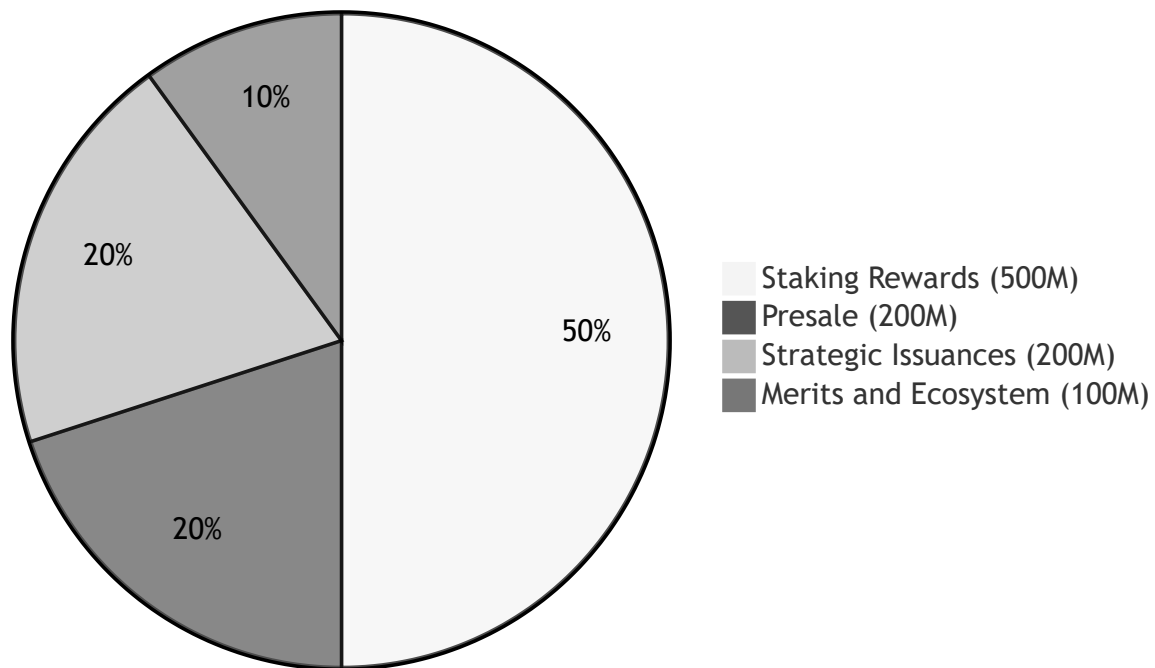
Consequences that must be respected throughout all protocol communication:

- A burn **may** be described as "returning minting capacity under MAX_SUPPLY", but never as "compensating the staking reserve".
- Any future policy on the ceilings requires express authorization by governance and cannot be presented as an automatic consequence of a burn.

2.3 Allocation limits

Program	Maximum limit	%	Nature
Presale	up to 200,000,000	20%	Genesis allocation (Section 5)
Merits and Ecosystem	up to 100,000,000	10%	Single contributions bucket
Staking Rewards	up to 500,000,000	50%	Immutable reserve (Section 6)
Strategic Issuances	up to 200,000,000	20%	Requires express DAO approval
Total	1,000,000,000	100%	

Maximum ZDAO allocation limits



This table expresses **maximum limits**, not fixed amounts that will be minted. Actual issuance depends on actual activity:

- **Presale and Merits** depend on what is actually raised and claimed. The Presale is denominated in **TOKEN ZERO** —an off-chain minting right— which is resolved at the Claim, at the participant's choice, into ZDAO or ZREC (Section 5.4); Merits allocations are converted directly into ZREC (Section 5.5). The ZDAO corresponding to a ZREC choice are never minted.
- **Staking Rewards** are backed by an **immutable reserve of 500,000,000 ZDAO**: only the staking contract can mint against it, and only when a reward has been legitimately accrued. The reserve is a ceiling, not an obligation (Section 6.3).
- **Strategic Issuances** are a ceiling that requires express DAO approval.

The 500,000,000 staking reserve is an **invariant**: it cannot be allocated to treasury, team, liquidity, Merits, campaigns, strategic issuances or any purpose other than the Official Staking System. ZDAO not needed for rewards are not minted.

The staking cap is cumulative. The staking contract keeps `stakingMinted`, a monotonic counter that never decreases through burning. Although burning ZDAO returns general capacity under `MAX_SUPPLY` (Section 2.2), it does **not** replenish the staking reserve: the staking contract can never mint more than 500M over its entire lifetime.

Merits and Ecosystem is a single bucket (abbreviated "Merits" throughout the document). It comprises the founding team, contributors, ambassadors and user onboarding campaigns. **There are no separate quotas by category.**

Strategic Issuances. Intended destinations, of an **indicative and non-binding** nature: market liquidity (150,000,000) and contingencies (50,000,000). Any issuance against this program requires a proposal

approved by the DAO and the timelock of Section 8. There is no automatic minting mechanism against this line item.

2.4 Fungibility

All issued ZDAO are fully fungible. The origin of their issuance alters neither the nature of the asset nor the rights it confers.

A technical consequence follows from this, which the protocol respects in its design: **an ERC-20 cannot distinguish the provenance of the tokens that make up a balance**. That is why no protocol rule depends on the origin of a specific ZDAO. The Presale choice between ZDAO and ZREC (Section 5.4), which does depend on the allocation, is resolved outside the token.

2.5 Functions

Governance. ZDAO confers voting rights over the configurable economic parameters, community proposals, the composition of the Protocol Committee, Strategic Issuances, the destination of the DAO Treasury's resources and protocol upgrades.

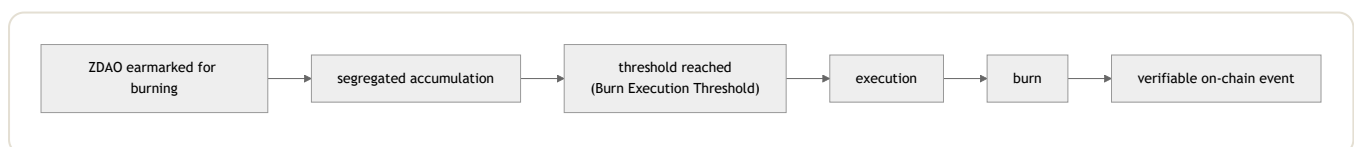
A precise distinction must be drawn: holders **decide by vote** the destination of the DAO Treasury. They hold no right to payment from it and do not participate in its distribution.

Incentive. ZDAO is the asset with which the protocol rewards participation in staking. Only those who have deposited ZREC receive these rewards. ZDAO does not represent energy production.

Relationship with protocol activity. ZERO DEX fees are distributed among the DAO Treasury, liquidity providers, the burn mechanism and infrastructure maintenance, according to the percentages in Section 7.

2.6 Burn mechanism

ZDAO contractually earmarked for burning **accumulate in a segregated and auditable manner** until a minimum execution threshold is reached.



Initial Burn Execution Threshold: 10,000 ZDAO. It is a parameter configurable by governance, not an economic constant.

The mechanism avoids hundreds of low-value burns, reduces operating cost and maintains on-chain traceability. Neither the DAO nor the Committee intervenes at the moment of execution: **the burn is triggered automatically by the fee-consolidation routine (a keeper or any user's transaction) when the segregated accumulation reaches the threshold**, and it emits a verifiable on-chain event. The segregated accumulation is held in a dedicated contract balance, separate from the circulating supply, until its execution.

In accordance with Section 2.2, ZDAO burned by this mechanism **reduce totalSupply and return minting capacity under MAX_SUPPLY**, but do **not** replenish the staking reserve (stakingMinted does not change).

3. ZERO REC (ZREC)

Energy asset of the protocol and the **only asset that can be deposited in the Official Staking System**. ERC-20 token on Base.

3.1 Standard and decimals

1 ZREC = 1 MWh = 1,000 kWh
0.1 ZREC = 100 kWh
0.01 ZREC = 10 kWh
0.001 ZREC = 1 kWh ← minimum representable unit

3 decimals. The correspondence with the energy unit is deliberate: the minimum unit of the token is exactly 1 kWh, which makes it possible to represent everything from residential microgeneration to industrial parks.

The difference in precision with ZDAO (8 decimals) is intentional and EVM-compatible. Contracts must normalize both scales with safe integer arithmetic and explicit rounding rules, above all in the calculation of staking rewards, where ZREC is deposited and ZDAO is issued.

ZREC **has no maximum supply**. It grows as new energy production is certified.

3.2 Fungibility: a single token

ZREC is **a single fungible asset**. There are no distinct tokens for Presale ZREC, Merits ZREC, energy-production ZREC or ZREC acquired on the market.

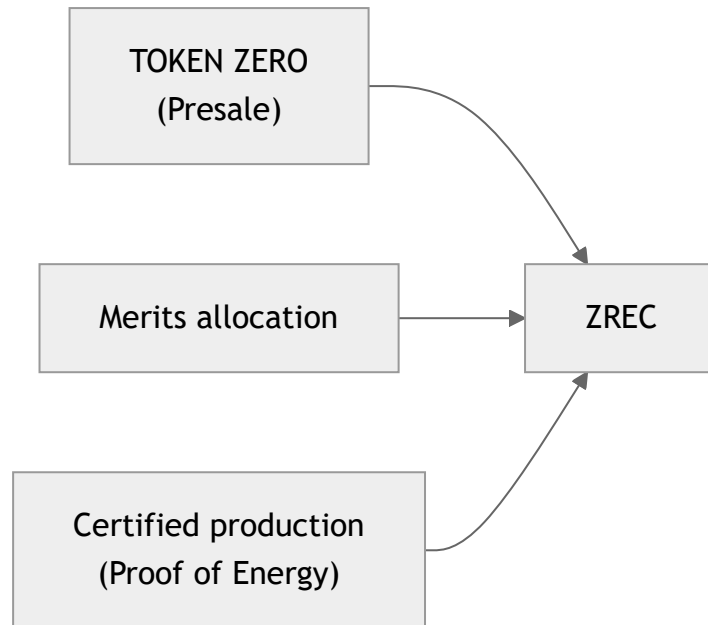
Differences in behavior apply **to the position or the contract where the asset is deposited, never to the token**. A ZREC released from a Merits position is exactly the same asset as any other free ZREC.

Provenance may be kept as historical traceability information, but it does not affect fungibility.

This principle allows the Merits lock to exist without fragmenting the asset or market liquidity.

3.3 Three origins

Origin	When	How	Energy backing
Presale ZREC	TGE + Claim window (6 months)	TOKEN ZERO minting right resolved into ZREC (Section 5.4)	No
Merits ZREC	TGE + Claim window (6 months)	Merits allocation resolved into ZREC (Section 5.5)	No
Energy ZREC	Permanent, after the window	Production certified through PoE	Yes



A. Presale ZREC

It is resolved **during the Claim**, in accordance with Section 5.4. A **TOKEN ZERO** minting right resolved into ZREC produces **free ZREC**: no lock and no mandatory stay in staking.

B. Merits ZREC

It allows the energy economy and staking to start before sufficient certified production exists. It is resolved **during the Claim**, in accordance with Section 5.5.

There is no mint-and-burn operation. When a Merits allocation is converted into ZREC, the corresponding ZDAO **are never minted on-chain**. The correct formulation is:

The conversion to ZREC extinguishes the right over the corresponding amount of ZDAO and enables the equivalent minting of ZREC, which is automatically deposited in staking with a special withdrawal schedule.

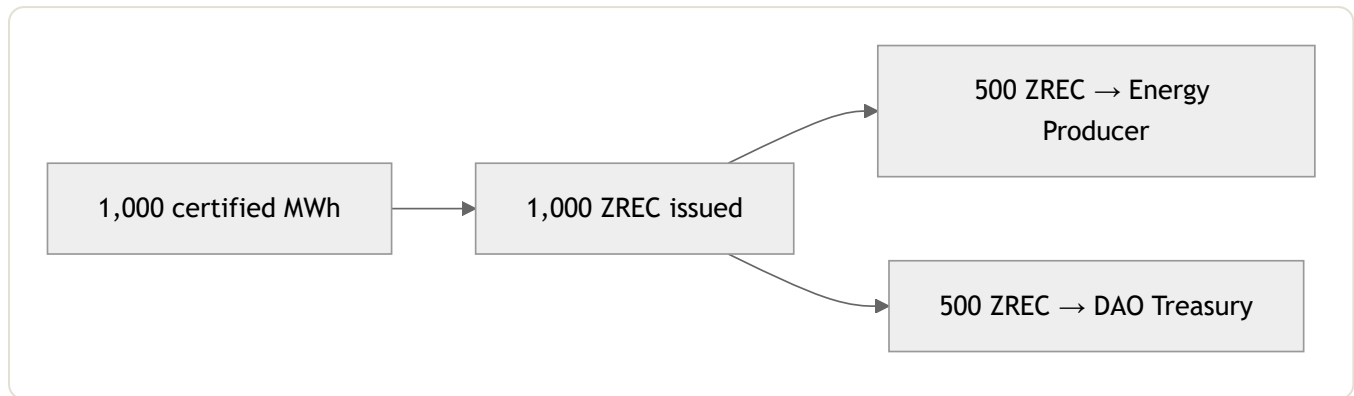


When the six-month window ends, this conversion path is **permanently and irreversibly disabled**.

C. Energy ZREC

The only ordinary source of issuance after the Genesis window. It cumulatively requires real production, validation by the Energy Oracles and finalization of the batch in accordance with Section 4.

Distribution of each issuance:



The 50/50 split is configurable by governance between 30% and 70% for the Energy Producer's share.

No initial per-producer issuance limit is established. Governance may introduce one in the future if the development of the protocol requires it.

3.4 Differentiated accounting

The protocol permanently and separately publishes:

- Presale ZREC issued.
- Merits ZREC issued.
- Energy ZREC issued.
- ZREC retired from circulation, by type.
- Percentage of the Genesis issuance offset (Section 3.6).

The separation is **exclusively for accounting purposes**: it affects the published aggregates, not the units or fungibility.

It is necessary because during the Genesis phase almost all ZREC in circulation will come from the Presale and from Merits and, therefore, **will not represent energy actually produced**. Publishing the aggregates separately makes it possible to know at all times what proportion of the energy asset is backed by real production.

3.5 Economic value

ZREC has no predetermined price or fixed parity with respect to ZDAO, any currency or any other asset. **The protocol does not establish, sustain or aspire to sustain any value for ZREC.** Its value is freely determined by the market.

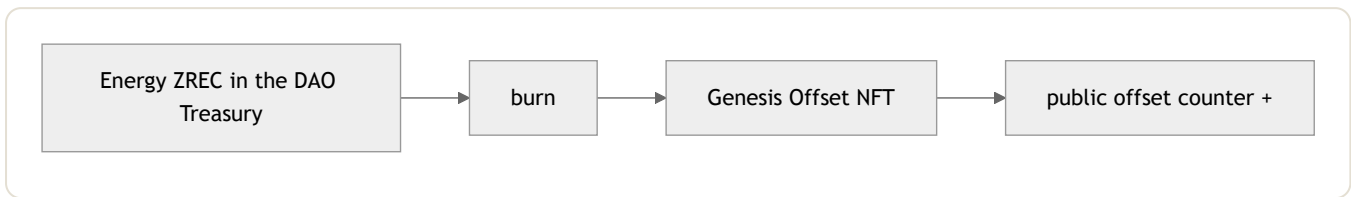
3.6 Genesis Offset Program

It progressively and verifiably attests that the Genesis issuance has been matched by real renewable production.

When the DAO Treasury receives Energy ZREC, it may allocate them to the program:

1. It selects ZREC accounted as Energy ZREC.

2. It permanently retires them from circulation by burning.
3. It issues the corresponding **Genesis Offset NFT**.
4. It increments the public cumulative offset counter.



Scope and limits. The program **does not modify, replace or individually back** the Genesis ZREC issued. It confers no right of exchange, refund or claim over energy. Its function is **accounting and demonstrative**: to publicly attest that the protocol has generated, certified and retired an amount of real renewable energy equivalent to a fraction of its Genesis issuance.

Single-use rule. A certified MWh cannot be counted twice. The Genesis Offset NFT is issued in the name of the DAO Treasury, is marked with that purpose and **is not transferable, nor can it be used by a third party as an environmental offset certificate**.

3.7 Energy Certification NFT

The protocol's energy certificate. Each NFT represents certified renewable energy permanently retired from circulation, and includes: unique identifier, Energy Producer and source installation, date, certified amount, Oracle validation reference, verifiable hash and **purpose** (third-party offset or Genesis Offset).

The purpose field determines the regime: Genesis Offset certificates are governed by Section 3.6 and are not transferable. The **Genesis Offset NFT** is, therefore, an instance of the Energy Certification NFT with purpose = `Genesis Offset`, not a distinct type of NFT.

Delimitation. ZREC certifies **renewable generation verified by the protocol**. The NFT certifies the **retirement** within the protocol. Neither of them automatically equates to an official Guarantee of Origin, an I-REC or any other regulated certificate, unless the corresponding integration and accreditation exist. This distinction must be maintained throughout all communication.

4. PROOF OF ENERGY AND ENERGY ORACLES

This section develops the layer that turns physical production into authorized ZREC issuance: what is certified, how the data is obtained, how it is validated and how it is grouped for on-chain registration.

4.1 What the protocol certifies

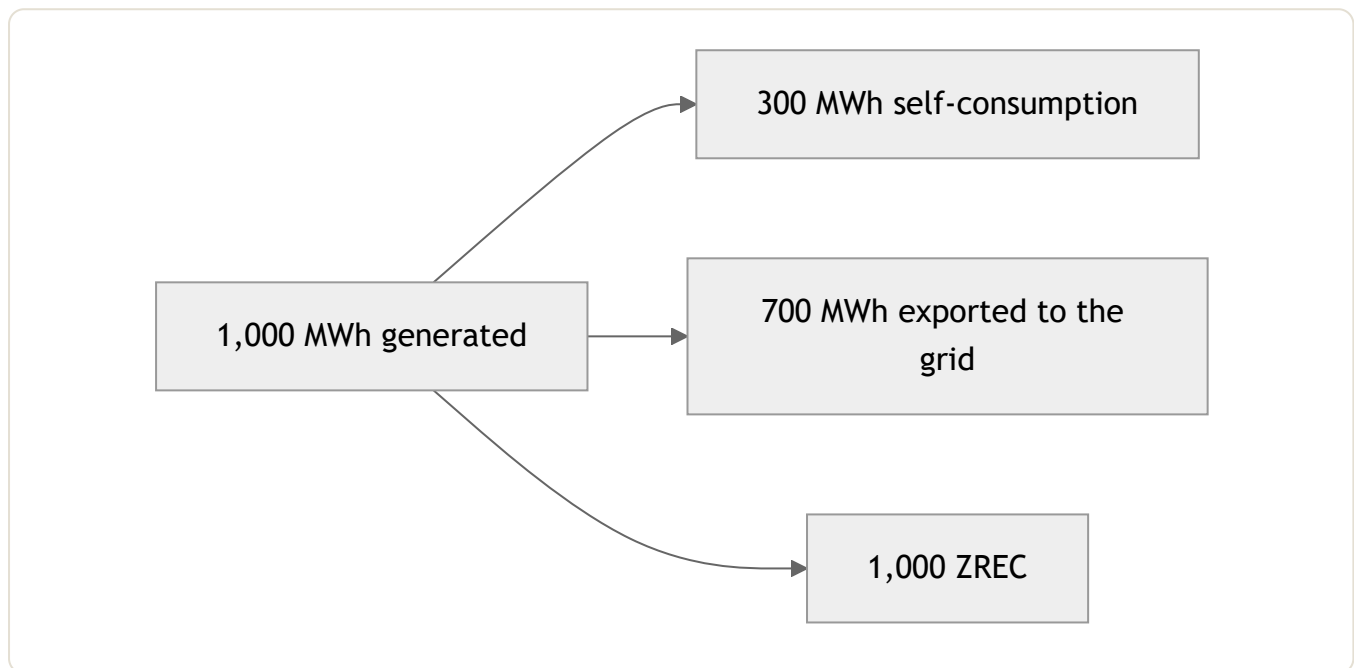
ZERO Energy rewards all renewable energy actually generated and validated, regardless of its subsequent destination —self-consumption, storage or export to the grid— and of whether or not that production is certified in other systems.

A producer who generates 1,000 MWh of renewable energy, self-consumes 300 and exports 700 to the grid has produced 1,000 MWh of renewable energy. If the protocol correctly verifies those 1,000 MWh at the

point of generation, they are potentially 1,000 ZREC.

In its initial phase, the protocol operates as a **self-sustaining, pilot-stage ecosystem**: through Proof of Energy, it rewards with ZREC the production information transmitted by producers, without discriminating by how they use that energy or whether they certify it elsewhere. Its objective is to demonstrate in practice that the protocol can solve the traceability problems of renewable energy. Interoperability with other certification systems will be addressed in the advanced phases of decentralization (Section 8.6).

A rule that excluded energy exported to the grid would be technically incorrect —the protocol measures generation, not consumption— and would penalize precisely the professional producers, whose main activity is injecting energy into the grid.



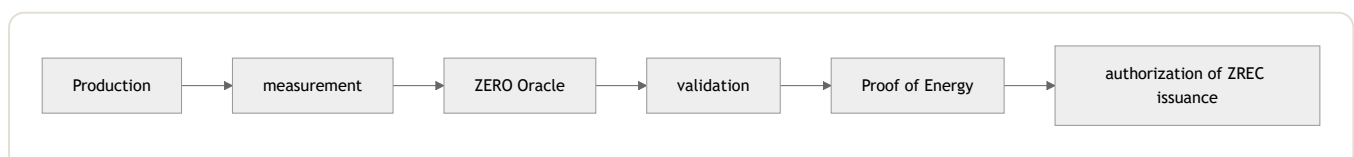
4.2 Objectives of the validation layer

- Verify real renewable production and guarantee the authenticity of the data.
- Prevent fraud, manipulation and unauthorized issuance.
- **Avoid double issuance**: each unit of energy is associated with a single ZREC issuance and, upon retirement, with a single certificate for a single purpose.
- Maintain an auditable record of all validations.

The non-duplication principle is cross-cutting and binds the rest of the document.

4.3 Oracle model and progressive decentralization

Initial model. During the initial phase, energy validation is carried out by an **oracle operated by ZERO Energy itself and authorized by it**.



The document **does not present the initial infrastructure as a decentralized oracle network**, because technically it is not one yet. Opening up to independent validators is a verifiable milestone of Section 8, not a statement of intent.

Evolution. The architecture is prepared to progressively admit multiple independent sources: smart meters, manufacturer APIs, inverters, SCADA, IoT systems, monitoring platforms, certifiers, external validators and grid operators.

It is not mandatory for all installations to have the same number or the same type of sources. The architecture allows **trust rules adapted to each installation typology**: a home with self-consumption and a photovoltaic park do not need the same level of redundancy.

Requirements for external validators. Their incorporation will follow security and maturity criteria: independence from the validated producer, technical capability, verifiable identity where necessary, availability, integrity and traceability of the data, cryptographic security, operational track record, absence of conflicts of interest and audited integration. The specific requirements that an external validator must meet are approved by the DAO by vote, with the timelock of Section 8.3.

The incorporation of a third party does not by itself constitute sufficient decentralization. The goal is to evolve toward truly independent validation sources that operate reliably in production for a sufficient period.

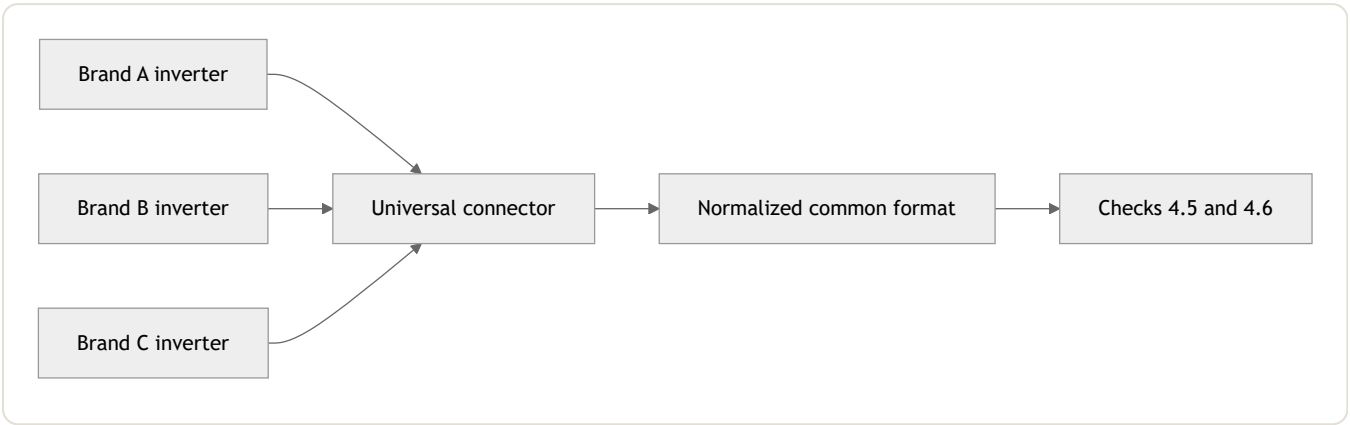
Uniqueness registry. The system maintains a registry of batch identifiers already processed. Any request that references an already registered identifier is automatically rejected, regardless of the oracle that submits it.

Delimitation between source and oracle. The source obtains and normalizes the data; the oracle verifies and signs it. A component that only reads data does not thereby acquire oracle status, however technically sophisticated it may be.

4.4 Data acquisition

Universal connector. The protocol integrates a connector capable of obtaining production data from installations of different manufacturers **without requiring additional hardware at the point of generation**: the holder selects the manufacturer of their inverter, authorizes access to their production data and the connector normalizes it to a common format.

This model substantially lowers the entry barrier for residential microgeneration, the most numerous segment of the distributed infrastructure.



Shift in the trust model. Not having a device at the measurement point does not reduce the risk: it shifts it from tampering with the meter toward the veracity of the declared installation and control of the manufacturer's account. That is why verification cannot rest on the data received and relies on the checks in 4.5 and 4.6.

4.5 Installation registration and verification

Before the first issuance, each installation is registered with: unique identifier, holder, location, **nominal power**, inverter manufacturer and model, and commissioning date.

Nominal power is not accepted on the producer's declaration. It is verified through three combined checks:

	Check	Content
A	Documentation	Technical datasheet, certificate or equivalent information on the installation
B	Identification	Verifiable link between the physical installation, the holder, the meter, the inverter and the monitoring system
C	Empirical validation	Comparison of recorded production against the nominal capacity and the physical characteristics

Empirical validation turns nominal power into a **physical coherence barrier**: an installation registered at 10 kWp that systematically declares 500 kW describes a physical impossibility, and the system can generate alerts, review, provisional suspension or a request for additional information.

Plausibility checks applicable to any production data:

- Coherence with the registered nominal power.
- Coherence with the irradiance corresponding to the location and date.
- Coherence with the weather conditions of the period.
- Coherence with the installation's own history, taking seasonality into account.
- Coherence with comparable installations in the same area and period.

These checks are **deterministic and reproducible** by any verifier that has the same input data.

Deliberate limitation. When anomaly detection incorporates machine learning models, their function is limited to **prioritizing cases for review**, with no capacity to authorize an issuance on their own. The authorization of a mint must be justifiable through verifiable rules: no component whose criterion is not reproducible can serve as the basis for an issuance.

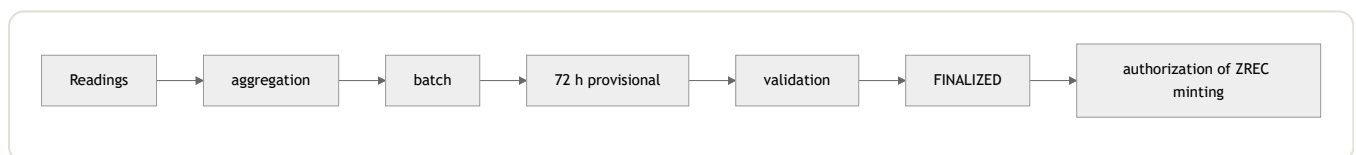
4.6 Energy batches

Production does not generate an on-chain transaction for each reading. Readings are aggregated into batches.

Batch closure. Initial configuration: **every 24 hours or upon reaching 1 MWh accumulated**, according to the configuration applicable to the installation and whichever occurs first. This hybrid criterion allows a solar home and a photovoltaic park to be managed with the same architecture. Parameter modifiable by governance.

Batch identifier. Each batch receives a unique identifier that **is the identifier in the uniqueness registry of Section 4.3**. A batch already processed cannot be processed again, however many times it is attempted. The system's idempotency derives from this: if a settlement fails and is retried, the registry prevents double minting.

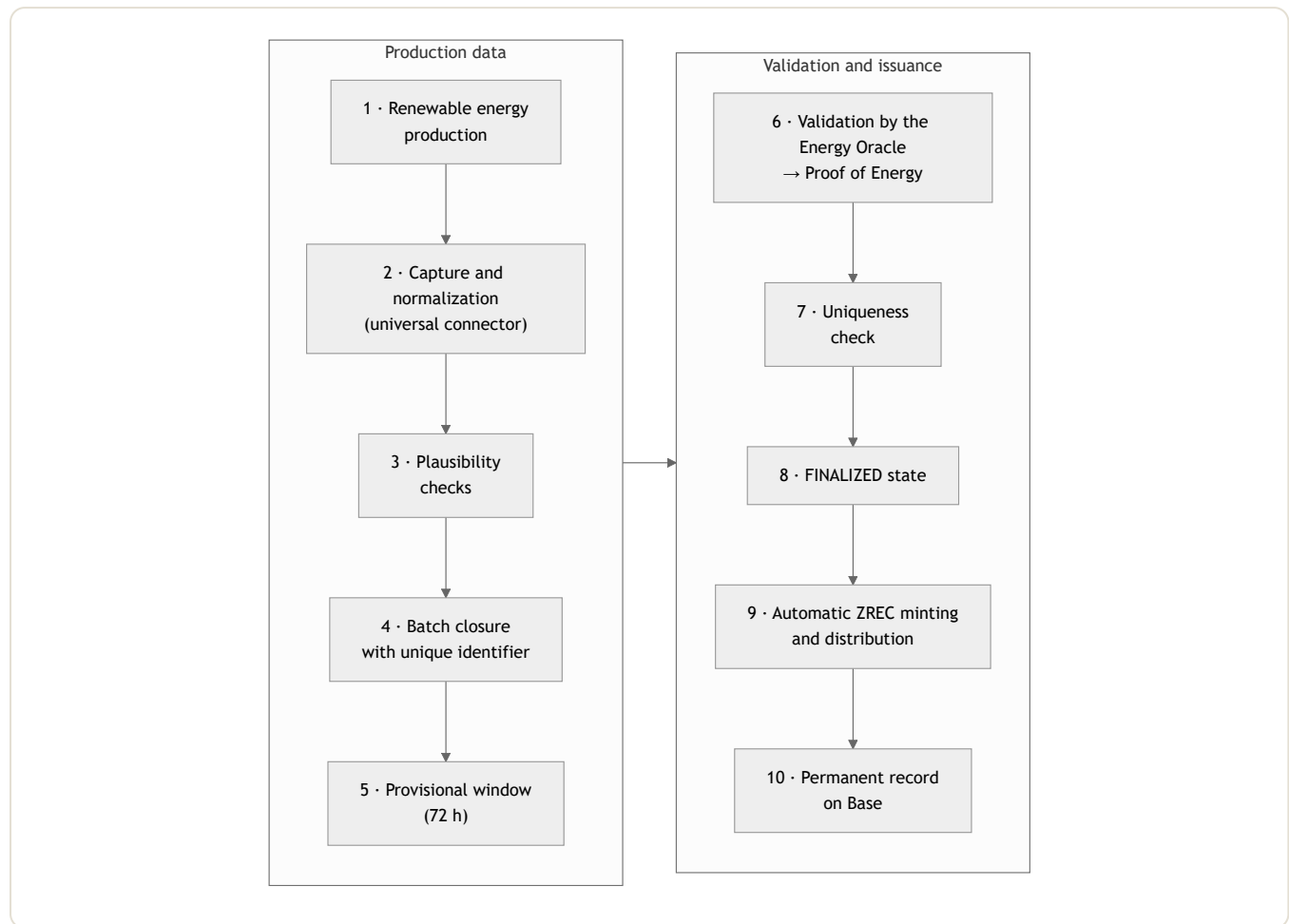
Provisional status: 72 hours. Once the batch is closed, a window opens during which data reconciliation, anomaly detection, cross-checking of sources, additional validations and technically justified corrections may be carried out.



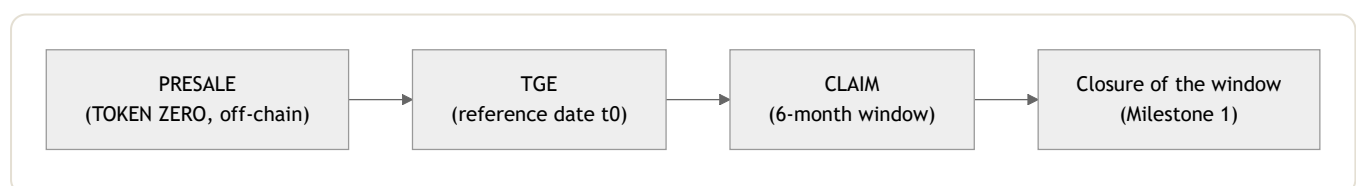
Upon reaching the **FINALIZED** state, the associated production is unequivocally identified as processed and cannot be reused in a second issuance. Minting is irreversible: the provisional window is the only mechanism that allows corrections to be incorporated. Governance may reduce its duration for high-trust producers or oracles.

Precision and residues. The conversion of accumulated energy into ZREC is performed with 3 decimals (minimum 1 kWh). **The non-representable fraction is carried over to the next batch**; under no circumstances is it discarded or rounded up. With a large number of low-power installations, systematic rounding in either direction produces a material accumulated deviation between certified energy and issued ZREC.

4.7 Complete validation flow



5. GENESIS PHASE: PRESALE, TGE AND CLAIM



5.1 Presale: TOKEN ZERO

Initial funding mechanism. Allocations are recorded **off-chain** under the Presale denomination **TOKEN ZERO**.

A TOKEN ZERO is a future minting right, not a minted token. What the participant acquires during the Presale is the right to mint, at the Claim, one ZDAO or one ZREC. Minting occurs solely through the Official Claim System after launch, which is where the right is resolved into ZDAO or ZREC. Separating funding from effective issuance enables a verifiable distribution without pre-minting.

Maximum allocation: 200,000,000 TOKEN ZERO (equivalent to 20% of MAX_SUPPLY). The percentages of each tranche are calculated **over those 200,000,000**, not over the global MAX_SUPPLY.

Tranche	% of the Presale	TOKEN ZERO	Price	Minimum investment	Amount raised
1	0% – 5%	10,000,000	0.10 USDC	1,000 USDC	1,000,000
2	5% – 15%	20,000,000	0.25 USDC	500 USDC	5,000,000
3	15% – 40%	50,000,000	0.50 USDC	250 USDC	25,000,000
4	40% – 70%	60,000,000	0.65 USDC	100 USDC	39,000,000
5	70% – 100%	60,000,000	0.75 USDC	10 USDC	45,000,000
Total		200,000,000	0.575 on average		115,000,000

The **Amount raised** column and the **average price of 0.575** are **theoretical maximums**: they assume full subscription of the five tranches. Since each tranche remains available until its units are exhausted, the actual amount raised may be lower.

The tranches operate in parallel. Each tranche has an independent allocation and remains available until its units are exhausted or until another closing condition established by the protocol occurs. **It is not necessary to exhaust one tranche to enable the next.** The minimum investment is the access criterion for each tranche.

Unallocated supply. Presale TOKEN ZERO that are not allocated are not minted and do not automatically pass to any other program.

The resources obtained are allocated to the development of the blockchain and energy infrastructure, the implementation and audit of the contracts, the onboarding of producers, the development of ZERO DEX and the growth of the ecosystem.

5.2 Token Generation Event (TGE) and Claim window

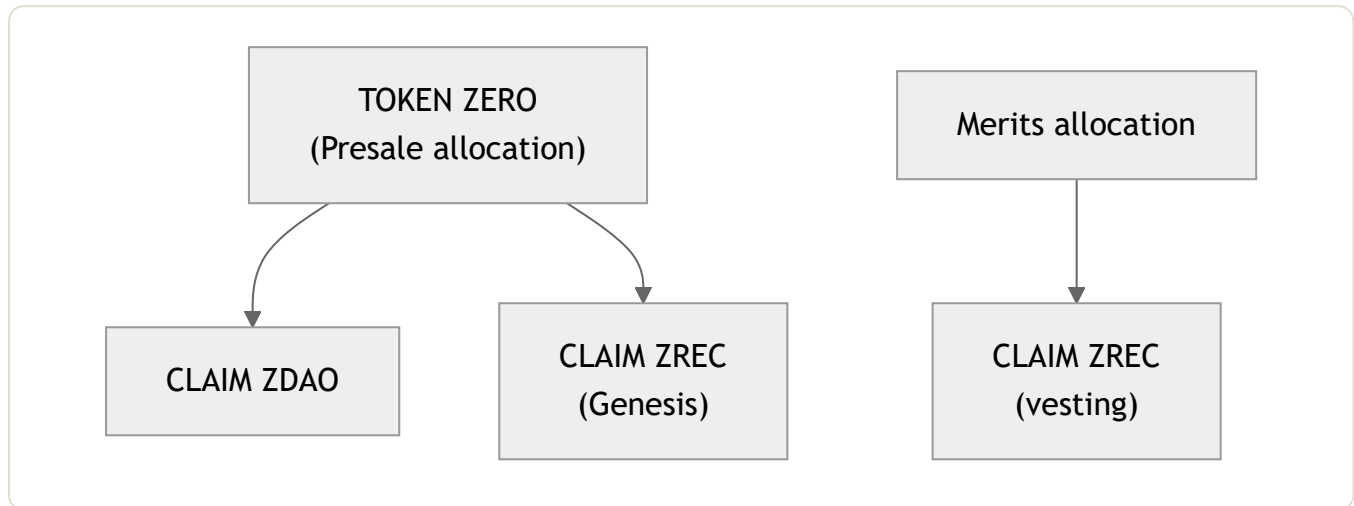
The TGE is the launch milestone that fixes the reference date and opens the Claim window. It does not mint or distribute assets by itself: minting occurs only when a participant claims their allocation.

- The TGE fixes the reference date (t_0) from which the Claim window and the Merits schedules are calculated.
- The Claim window lasts **six (6) months** from the TGE.
- Presale participants exercise their **TOKEN ZERO** minting right by choosing, at the Claim, between ZDAO and ZREC (both without vesting).
- Merits participants convert their allocations into ZREC with a vesting schedule (Section 5.5).
- The staking positions corresponding to Merits ZREC are created automatically.

When the window closes, the possibility of claiming a Genesis allocation also closes. **Allocations not claimed within the window lose the minting right**: those tokens are never minted.

5.3 Claim options

Each Genesis allocation admits, at the moment of the Claim, a single, one-time choice that definitively consumes the allocation. Only Presale allocations (TOKEN ZERO) involve a choice; Merits allocations are resolved directly into ZREC (Section 5.5).



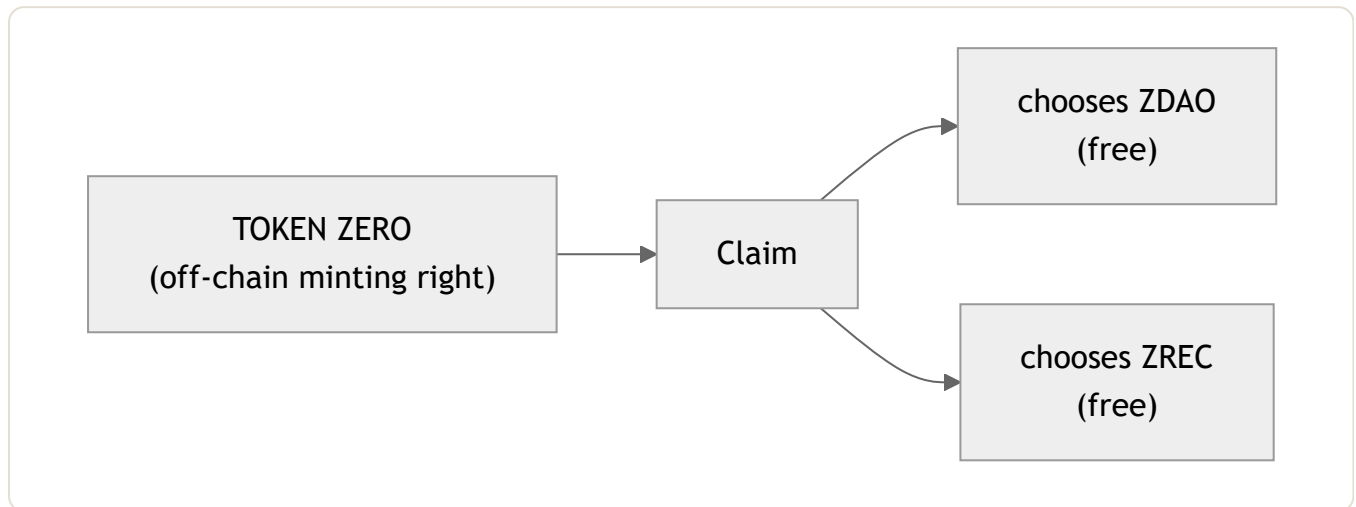
Once an option is executed, the allocation is definitively consumed. Consequently:

- A second Claim cannot be made on the same allocation.
- It cannot be duplicated using different wallets.
- It cannot be exercised with tokens acquired on secondary markets.

Since ZDAO and ZREC are ordinary fungible tokens (Sections 2.4 and 3.2), an ERC-20 cannot distinguish the provenance of the tokens in a balance. That is why the choice is resolved in the off-chain allocation registry during the Claim, never in the token. This eliminates by design the vector *wallet A claims* → *transfers to wallet B* → *wallet B tries to claim again*: the operation is impossible because the allocation was consumed in the initial Claim.

5.4 Presale branch: ZDAO or ZREC

A **TOKEN ZERO** (the Presale minting right) is resolved, at the Claim, into **ZDAO or ZREC**, at the participant's choice. Both options are **free of vesting**: no lock and no mandatory stay in staking.



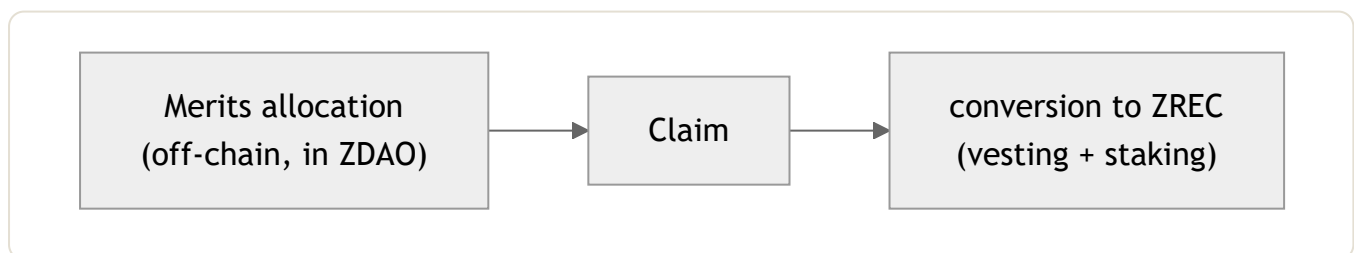
Ratio: **1 TOKEN ZERO = 1 ZDAO or 1 ZREC**. There is no burn: the ZDAO of a ZREC choice are never minted.

- **ZDAO**: minted on-chain, completely free, fungible and transferable, with no lock.
- **ZREC**: minted on-chain, completely free, with no vesting, no lock and no minimum stay in staking. The participant may hold it, transfer it, sell it, swap it, provide it as liquidity or voluntarily deposit it in staking under the general withdrawal rules.

This path gives Presale participants direct entry into the protocol's energy economy and into ZREC staking.

5.5 Merits branch: ZREC with vesting

Merits allocations are recorded off-chain, denominated in ZDAO. At the Claim, the participant **converts the allocation into ZREC** (there is no ZDAO option). The ZREC is minted and **automatically deposited in staking**, where it remains generating ZDAO under a special withdrawal schedule: the Merits **vesting**.



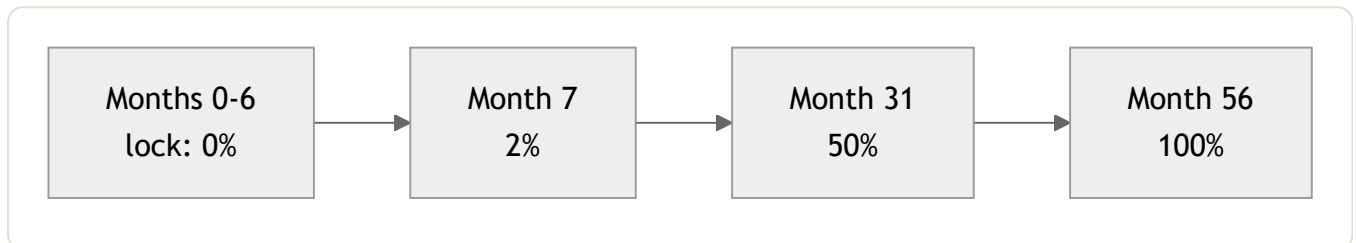
Ratio: **1 ZDAO allocated through Merits = 1 ZREC**. There is no burn: the ZDAO are never minted.

Vesting is not a separate contract. The time restrictions are applied **directly through the withdrawal rules of the position within the Official Staking System**. The correct formulation is "**Merits ZREC → staking with a special withdrawal schedule (vesting)**".

The ZREC remain **in staking throughout the entire schedule** and generate ZDAO from day one. The ZDAO generated as rewards are **free and immediately transferable** (Section 6.5): the lock applies only to the deposited ZREC principal.

Availability schedule (vesting)

Phase	Duration	Availability
Initial lock (cliff)	6 months	0%
Progressive release	50 months	2% monthly of the initial allocation
Total	56 months	100%



During the first six months, no principal may be withdrawn, the deposited ZREC may not be sold or transferred, and no mechanism intended to circumvent the lock may be used. The position participates in staking from day one.

Example: initial allocation of 100,000 ZREC

Months 0-6 0 ZREC available
Month 7 2,000 ZREC available
Month 8 4,000 ZREC available
Month 31 50,000 ZREC available
Month 56 100,000 ZREC available

Each monthly tranche that reaches its date may be withdrawn. **Once withdrawn, the ZREC is completely free** and equivalent to any other. The restriction affects exclusively the part of the position that has not yet reached its availability.

5.6 Official Claim System

Prior consolidation. Before the Claim opens, the allocation registry is closed, a Merkle tree is generated with all allocations and their conditions, its root is published on-chain and the complete set is published for independent verification. From that moment, the set is immutable and any participant can verify their allocation without depending on the protocol.

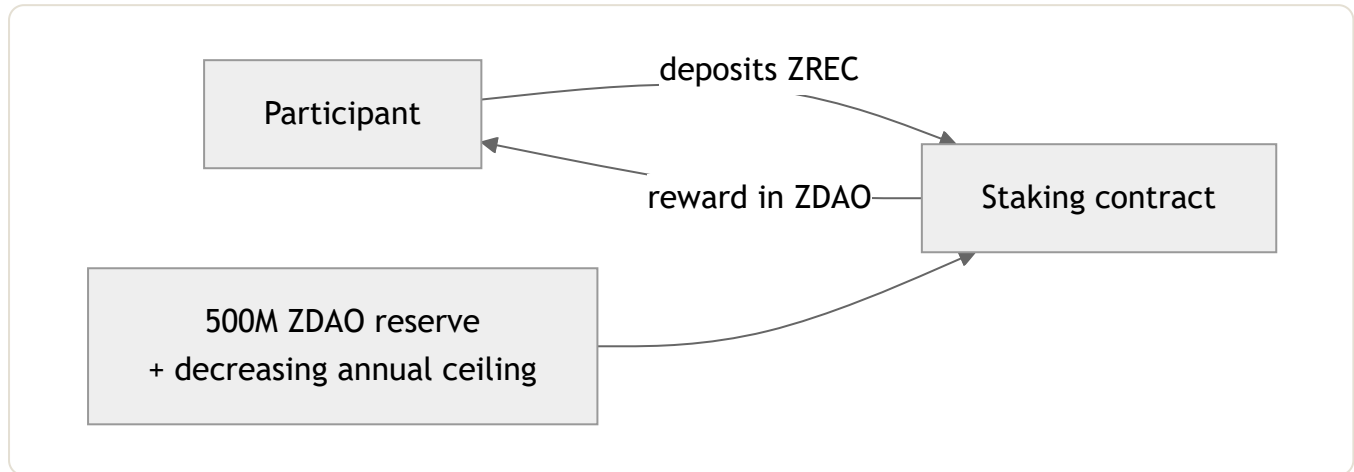
Execution. The participant connects a Base-compatible wallet, obtains their proof of inclusion and submits it to the contract, which verifies the proof against the published root, checks that the allocation has not been consumed, resolves the corresponding branch and marks the allocation as permanently consumed.



This model is preferable to an authorization issued by a server: the published root fixes the set of allocations immutably and auditably **before a single asset is minted**, and no external infrastructure retains the capacity to authorize minting. Any modification of the root requires DAO approval and the timelock of Section 8.

6. OFFICIAL STAKING SYSTEM

ZREC is deposited and **ZDAO** is received. It is the only mechanism through which the 500,000,000 ZDAO of the rewards reserve can be minted, within the limit of the decreasing curve (Section 6.3).



6.1 Technical basis

The Official Staking System is a **proprietary development of ZERO Energy, with its own rules**: a decreasing rewards model in which the annual rate is 100% in the first year and halves each year, with an issuance ceiling of 250,000,000 ZDAO in the first year that also halves each year.

The asset that is deposited (ZREC) and the one received as a reward (ZDAO) are different, which requires explicitly defining the relationship between the two and the normalization between their decimal scales (3 in ZREC, 8 in ZDAO).

6.2 Rewards model

The system **does not admit any interpretation as a guaranteed fixed return**. The reward accrues **linearly over time**, with an annual rate that **halves each year**: a position held for a full year accrues **100% of its principal**; a second full year, **50%** of the principal; a third, **25%**; and so on. The reward is expressed in **ZDAO per deposited ZREC**.

Time component (position age)

The reward accrues **linearly** within each year of staking. The annual reward rate, according to the year of staking, is:

$$r_n = \frac{100\%}{2^{n-1}}$$

Staking year	Annual rate
Year 1	100%
Year 2	50%

Staking year	Annual rate
Year 3	25%
Year 4	12.5%
Subsequent	halves each year

A position accrues the rate of its current year **in proportion to the time actually deposited within that year**. The rate is flat within each year: there is no preferential weighting of early or late holding periods.

Accrual

The total reward is the sum, for each year of staking, of that year's rate applied to the time deposited in that year:

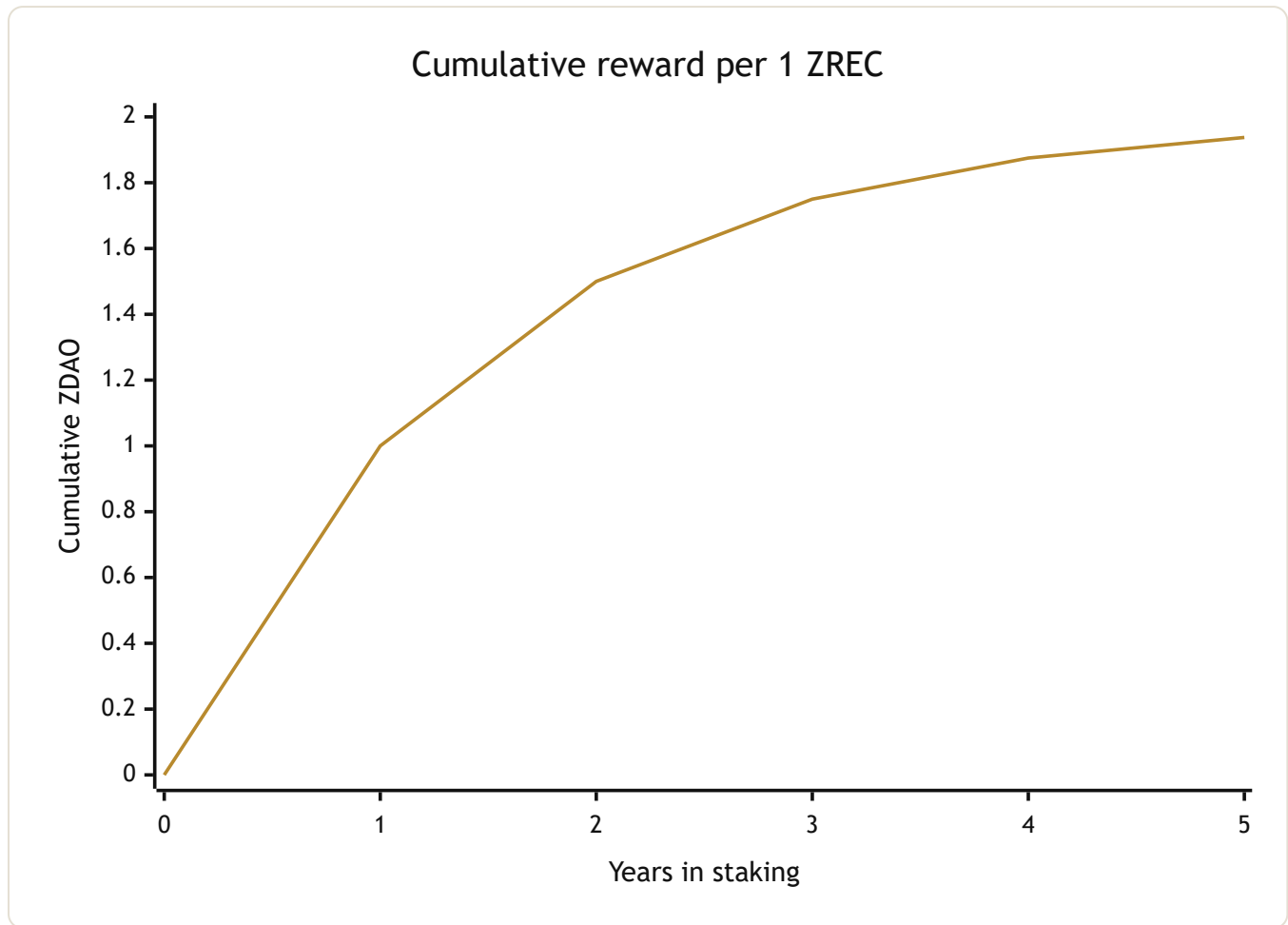
$$reward = amount \cdot \sum_{n \geq 1} r_n \cdot \frac{t_n}{Y}$$

where $Y = 365$ days and t_n is the portion of year n during which the position was deposited (in days, with a maximum of Y). Specifically:

- $T = 0 \rightarrow reward = 0$.
- $T = 365 \text{ days} \rightarrow reward = amount$ (**100% of the principal**).
- $T = 2 \text{ years} \rightarrow reward = 1.5 \cdot amount$ (**150%**, i.e., 100% + 50%).
- $T = 3 \text{ years} \rightarrow reward = 1.75 \cdot amount$ (**175%**).
- A half-year position $\rightarrow reward = 0.5 \cdot amount$ (**50%**).

Accrual is therefore **piecewise linear**: a flat annual rate that halves at each year boundary.

Cumulative reward per 1 deposited ZREC, in ZDAO, before applying the ceilings of Section 6.3:



Why this form. Linearity makes the reward transparent and predictable at any moment, with no normalization constants or slope parameters to adjust. The halving schedule (100% → 50% → 25% ...) follows **the same pattern** as the annual issuance ceiling of Section 6.3, so that the reward rate and the issuance limits decrease together.

No automatic compounding. There is no separate growth factor or compound interest: linear accrual alone determines the entire reward.

Only the ZDAO actually accrued are minted, within the reserve and the annual ceilings of Section 6.3. ZDAO not needed for rewards are simply not minted.

Scope of the figures. The rates in this section describe the contract's calculation in ZDAO per ZREC. They are neither a return in any currency nor a guarantee: the economic value of a reward depends on the price ratio between ZDAO and ZREC, which is set by the market, and on the ceilings of Section 6.3, which may reduce the accrual of all positions.

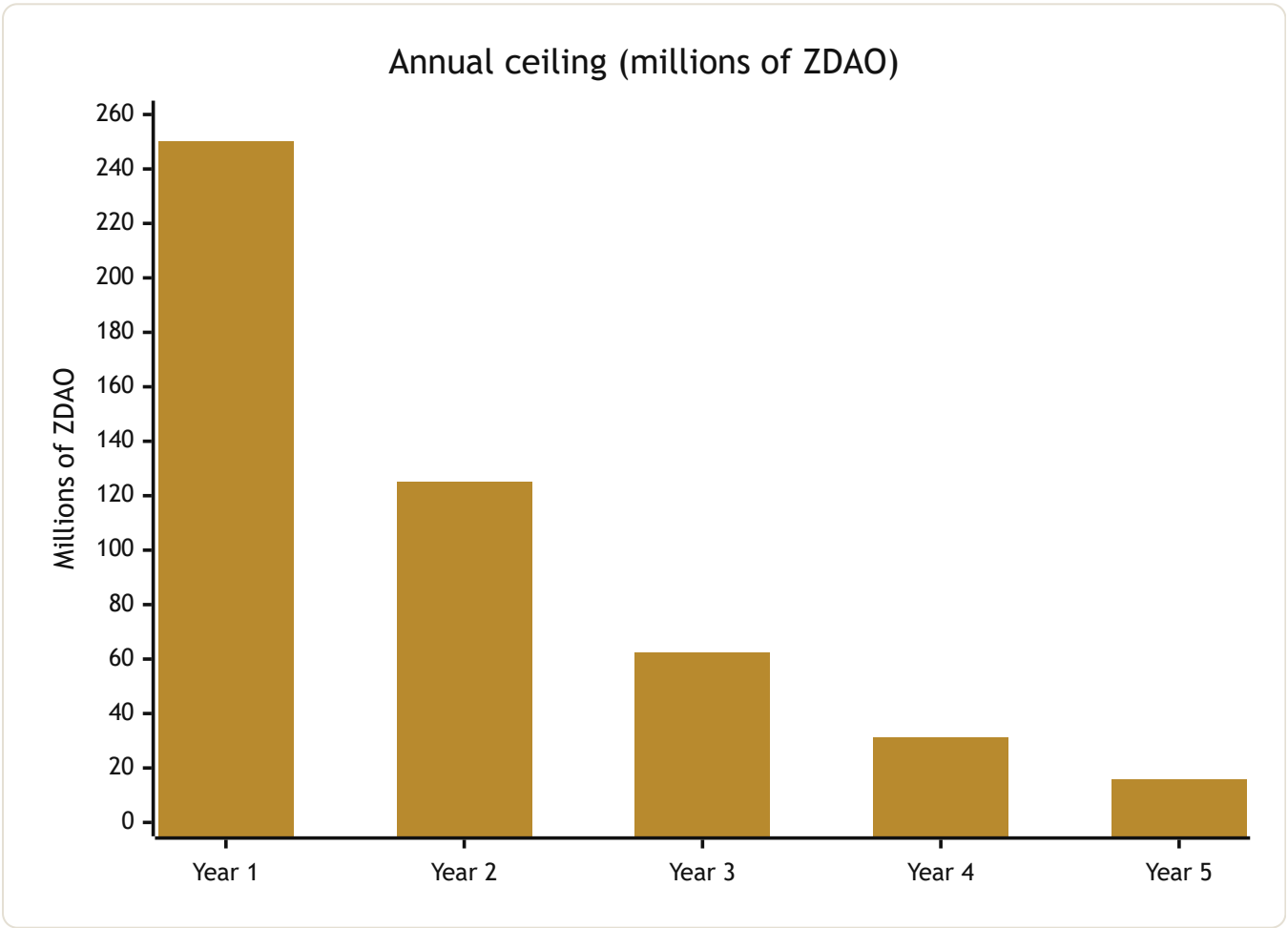
6.3 Issuance limits

Two limits operate cumulatively on the result of the formulas above, together with the MAX_SUPPLY ceiling.

1. Absolute staking reserve — invariant. 500,000,000 ZDAO. It cannot be allocated to treasury, team, liquidity, Merits, campaigns, strategic issuances or any purpose other than the Official Staking System. Only the staking contract can mint against it, and only when a reward has been legitimately accrued.

2. Decreasing maximum curve. Issuance ceiling per year, as an upper limit:

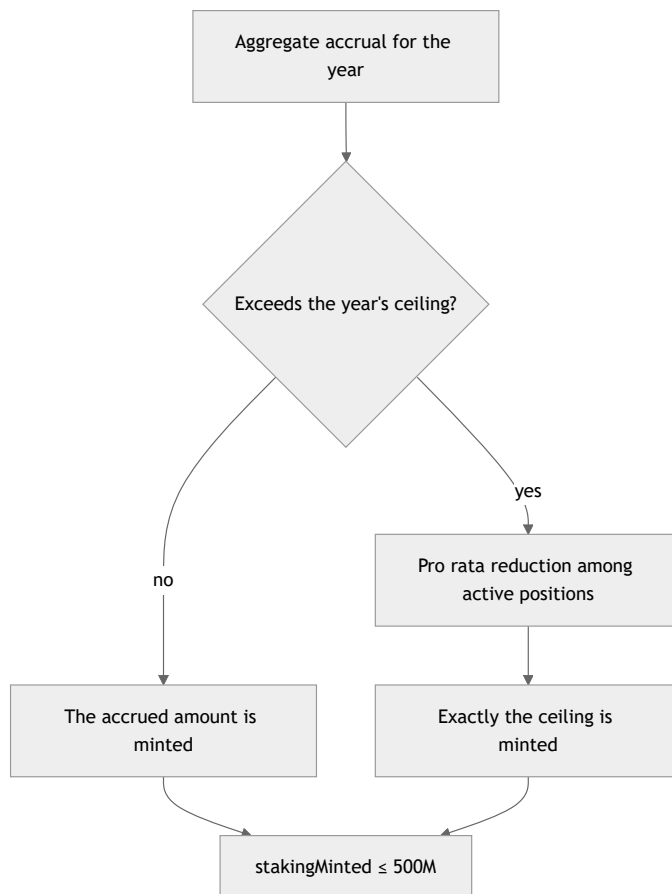
Year	Maximum issuance
Year 1	up to 250,000,000 ZDAO
Year 2	up to 125,000,000 ZDAO
Year 3	up to 62,500,000 ZDAO
Year 4	up to 31,250,000 ZDAO
Subsequent	halves each year



The first-year ceiling is 250,000,000 ZDAO and halves each subsequent year. These values are **maximum ceilings, not issuance obligations**: if the rewards accrued in a year fall below the available ceiling, the difference **is not minted automatically**. The document must not describe these amounts as figures that will necessarily enter circulation.

Why the ceiling is necessary. ZREC grows without limit as certified production increases. A percentage rate applied to a growing deposited base produces growing issuance; without a ceiling, the greater the protocol's energy success, the sooner the rewards reserve would be exhausted. The decreasing curve bounds that risk.

Mechanism. The staking contract accounts for the ZDAO minted in each year against the curve. If the aggregate accrual of a year were to exceed that year's ceiling, the accrual is reduced pro rata among the active positions, so that the year's total never exceeds the ceiling. The unused portion of a year's ceiling does not carry over to the next.



Only the ZDAO actually accrued are minted (and within both ceilings). Issuance thus follows real usage — participation in staking and circulating supply—, bounded by the reserve, the decreasing curve and MAX_SUPPLY.

6.4 Position types

The system uses **individualized positions**, each with its own amount, entry date, age, accumulated rewards, type, status and withdrawal availability. Each contribution constitutes an independent position with its own age computation.

	Free staking	Merits staking
Applicable to	Presale ZREC, Energy ZREC, market ZREC	ZREC from the Merits conversion
Entry	Voluntary	Automatic during the Claim
Lock	No	Initial 6 months

	Free staking	Merits staking
Availability	General withdrawal rules	2% monthly for 50 months (vesting)
Rewards	General model	General model, identical

Both types use the same ZREC. No distinct versions of the token are created to represent its provenance (Section 3.2). The difference lies **exclusively in the principal withdrawal rules**.

6.5 Rewards of Merits positions

Merits ZREC **generate rewards from the start of their position, including the six-month lock period**, under the same general economic model. There is no differentiated formula.

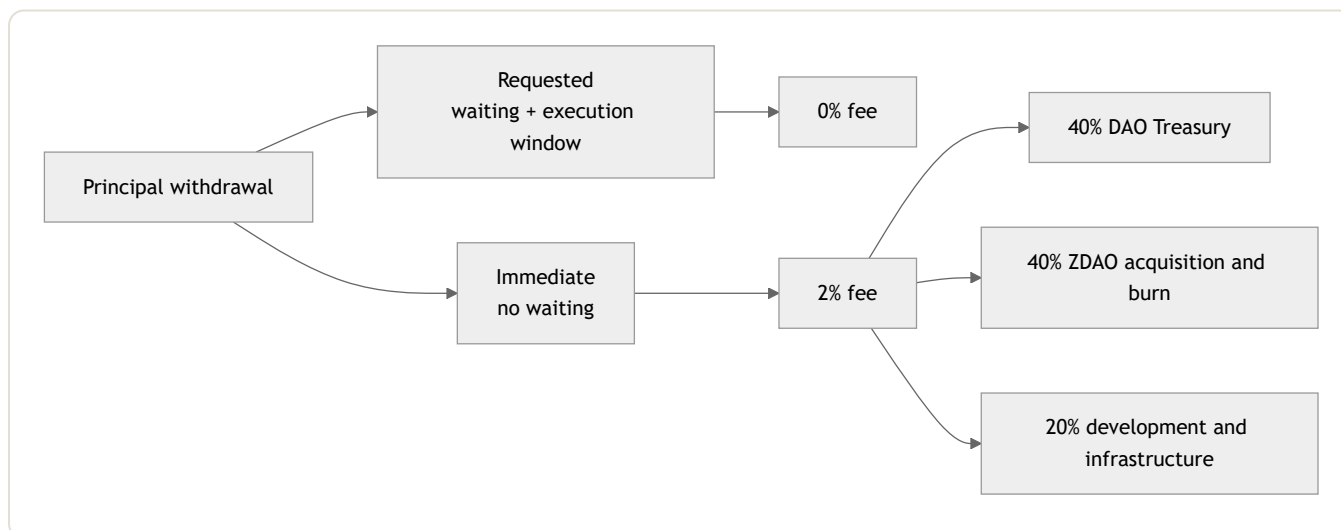
The ZDAO generated as rewards:

- come exclusively from the staking reserve, not from the Merits Program;
- **do not inherit the lock** of the ZREC that generated them;
- are free as soon as they are distributed.

6.6 Withdrawal

Two modalities for free positions and for the available portion of Merits positions:

Modality	Procedure	Fee
Requested	Waiting period + execution window	0%
Immediate	No waiting period	2% of the principal withdrawn



Immediate withdrawal fee: 2% initially on the principal withdrawn. This is the initial value, not a permanent value: the DAO may modify it by vote within the **range of 0.50% to 5.00%**.

Initial fee split, modifiable by governance within the limits of the protocol:

Destination	Initial
DAO Treasury	40%
ZDAO acquisition and burn	40%
Development and infrastructure	20%
Liquidity providers	0%

This fee is independent of the ZERO DEX fees and does not necessarily follow their split. The portion allocated to acquisition and burn is accumulated and executed according to the Burn Execution Threshold of Section 2.6.

Merits positions. Immediate withdrawal does not allow the lock to be circumvented: during the first six months, no principal may be withdrawn through either route. Thereafter, only the amount already available according to the progressive release of 2% monthly over 50 months may be withdrawn (Section 5.5).

7. ZERO DEX

7.1 Nature

ZERO DEX is the main Dashboard and the decentralized interface of the ecosystem, not merely an exchange or a set of liquidity pools. It is the layer through which users interact with the protocol's on-chain infrastructure.

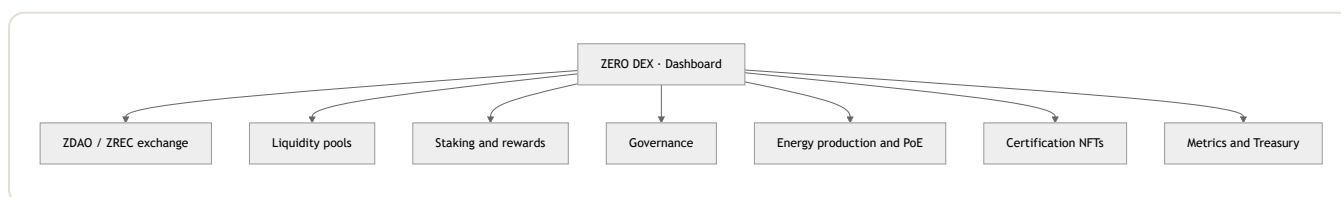
The following distinction must be made expressly:

ZERO DEX = Dashboard and interface of the ecosystem
 Contracts, pools, routers, liquidity protocols = underlying technology layer

ZERO DEX may interact with proprietary, external or aggregated infrastructure.

7.2 Functionalities

They are integrated progressively: exchange of ZDAO, ZREC and compatible assets; access to liquidity pools; ZREC staking and position management; collection of rewards; interaction with governance; energy production and Proof of Energy information; visualization of ZREC and of Energy Certification NFTs; producer and installation information; protocol metrics and Treasury information; DeFi functionalities and future applications built on ZERO Energy.



Restriction. The market operates exclusively on a spot basis. No trading with leverage, margin or derivative instruments on the ecosystem's assets is offered. Incorporating any functionality of that nature requires an express DAO decision and a prior review of the architecture.

7.3 Fees

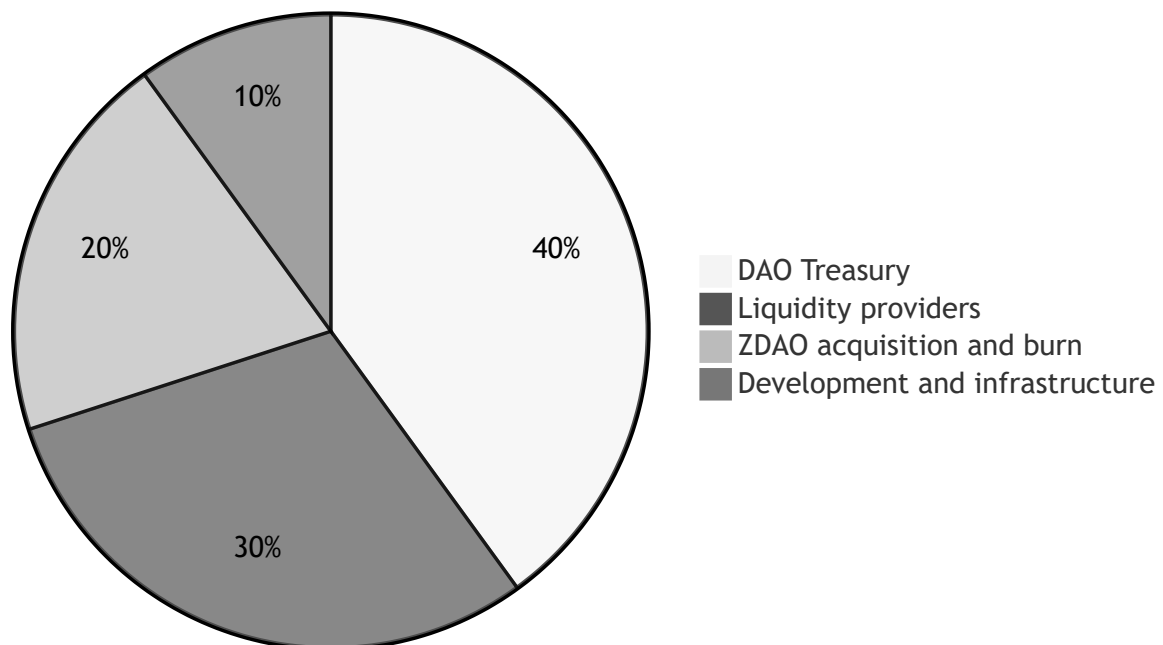
Operations generate a **fee on the traded volume**, which is collected in the input asset and consolidated periodically.

Fee: 0.30% of volume (initial), configurable by governance in the range 0.05% – 1.00% (Section 11.2).

The fee is distributed among the following destinations:

Destination	Initial	Configurable range
DAO Treasury	40%	20% – 50%
Liquidity providers	30%	20% – 50%
ZDAO acquisition and burn	20%	0% – 30%
Development and infrastructure	10%	0% – 20%

Initial ZERO DEX fee split

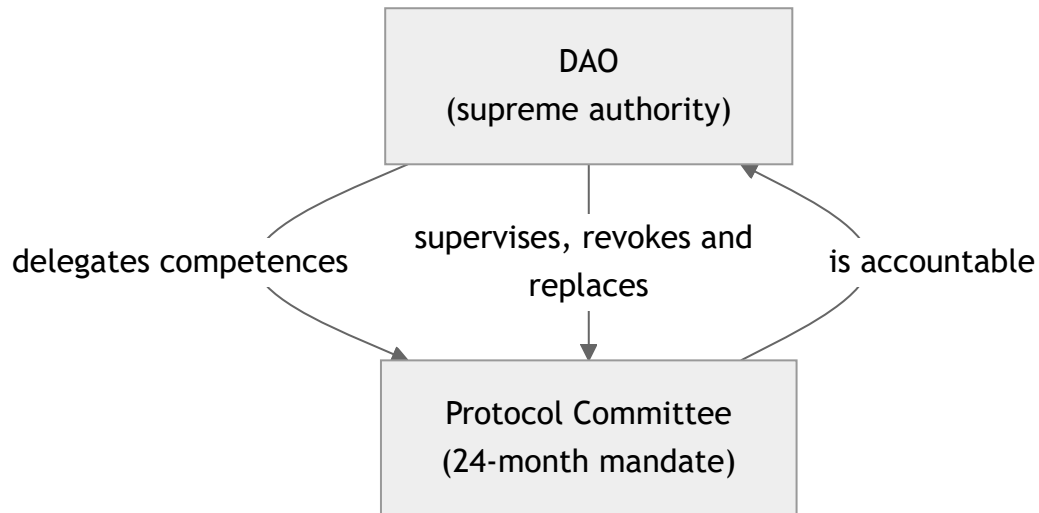


The sum must at all times equal 100%; the contract rejects any configuration that does not comply with this. The fraction allocated to burning is accumulated and executed according to the Burn Execution Threshold of Section 2.6.

8. GOVERNANCE AND PROGRESSIVE DECENTRALIZATION

8.1 DAO and Protocol Committee

The **DAO is the supreme governance authority**. The **Protocol Committee** is an operational body with delegated competences, **not an authority above the DAO**.



Initial mandate of the Committee: 24 months. One year is insufficient to manage energy infrastructure; periods of four or five years concentrate too much power. Twenty-four months allow for both work and accountability.

The Committee is subject to renewal, early revocation through the governance mechanisms, replacement for justified cause, transparency obligations, accountability, limits on its competences and supervision by the DAO.

The Committee cannot mint tokens, modify parameters outside their range or dispose of the DAO Treasury without prior approval.

8.2 Administrative permissions

The protocol recognizes exclusively the following permissions. No other permission exists over the core contracts.

Permission	Holder	Scope	Lifecycle
Reward minting	Staking contract	500M reserve, bounded by the decreasing curve	Permanent
Minting via Claim	Claim contract	Allocations from the published root	Extinguished when the window closes

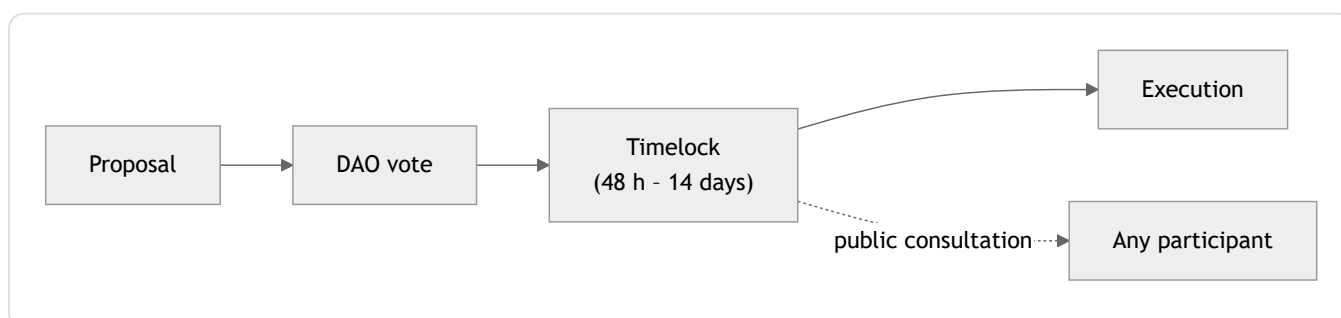
Permission	Holder	Scope	Lifecycle
Genesis ZREC minting	Claim contract	Presale (TOKEN ZERO) and Merits allocations resolved into ZREC	Extinguished when the window closes
Energy ZREC minting	Energy Issuance contract	Finalized batches	Permanent
Oracle operation	ZERO Energy	Signing of validations	Opened to third parties (Milestone 3)
Strategic minting	Governance	Strategic Issuances	Subject to the supply invariant
Parameter configuration	Governance	Ranges of Section 11	Permanent
Contract upgrades	Governance	Core contracts	Renounced (Milestone 5)
Emergency pause	Protocol Committee	Temporary suspension	Renounced (Milestone 5)

No minting permission rests with an address controlled by individuals. Permissions are held by contracts, whose behavior is verifiable.

8.3 Timelock

Every execution arising from the governance contract is subject to a mandatory waiting period, during which the pending transaction can be publicly consulted.

Operation	Minimum period
Parameters within range	48 hours
DAO Treasury operations	72 hours
Strategic Issuances	7 days
Modification of the oracle set	7 days
Upgrade of core contracts	14 days



Its purpose is to guarantee that no decision affecting the assets or the logic of the protocol can be executed without participants having time to learn of it and react.

8.4 Emergency pause

The Committee may temporarily suspend operations in the event of a security incident, with strict limits: **it does not allow minting, transferring or disposing of any asset**; maximum duration of 72 hours with automatic lifting; extension subject to DAO approval; public justification within 24 hours.

8.5 Contract upgrades

The core contracts are deployed with upgrade capability during the initial phases, in order to correct defects detected in audit or in production. Every upgrade requires DAO approval, a 14-day timelock, a prior audit and publication of the proposed code throughout the entire waiting period.

No upgrade may modify an invariant of Section 11. The invariants are implemented so that their verification does not depend on upgradable code. The upgrade capability is extinguished at Milestone 5.

8.6 Progressive decentralization by milestones

Decentralization is not tied to calendar dates. A protocol does not become decentralized because a date arrives. Each milestone is defined through **objective and verifiable conditions**, and some conditions must be met simultaneously, so that no one can artificially accelerate the process in order to gain power or unlock a capability.

Variables that may be considered jointly: number of independent producers, certified energy, diversity of data sources, number and distribution of validators, maturity of the Energy Oracles, TVL, maturity and audits of the contracts, activity and distribution of participation in governance, and operational security.

The specific criteria for each milestone are approved by the DAO by vote, with the timelock of Section 8.3, as the protocol evolves.

Each milestone entails an **irreversible renunciation** of a control capability: once executed, it cannot be reversed by any mechanism, including contract upgrades.

Milestone	Renunciation	Verification
1. Closure of the Genesis window	The Claim is disabled	The functions revert unconditionally
2. Consolidation of issuance	ZDAO minting permissions are extinguished, except staking and strategic	Public list of authorized addresses
3. Independence of validation	ZERO Energy ceases to control energy validation	Public list of oracles and their accredited independence
4. Ecosystem maturity	Criteria defined by governance	Published metrics
5. Immutability	Contract upgrades and the emergency pause are renounced	Administrative address with no capacity to act

Milestone	Renunciation	Verification
6. Full governance	The Committee loses all capacity for initiative	Proposals originate only from the governance contract

Until **Milestone 3** is executed, energy validation depends on infrastructure operated by ZERO Energy. The document states this expressly in Section 4.3 rather than presenting as decentralized a mechanism that is not yet so.

8.7 Publication of status

The protocol publishes, permanently and in a consultable manner: the milestone reached and the transaction of each renunciation executed; the list of addresses with an active minting permission; the pending transactions in the timelock; the composition of the Committee and of the oracle set, indicating their operator; `totalSupply`, the remaining issuance capacity and the status of the staking reserve (`stakingMinted`); the number of registered installations and the cumulative certified energy; and the percentage of the Genesis issuance offset.

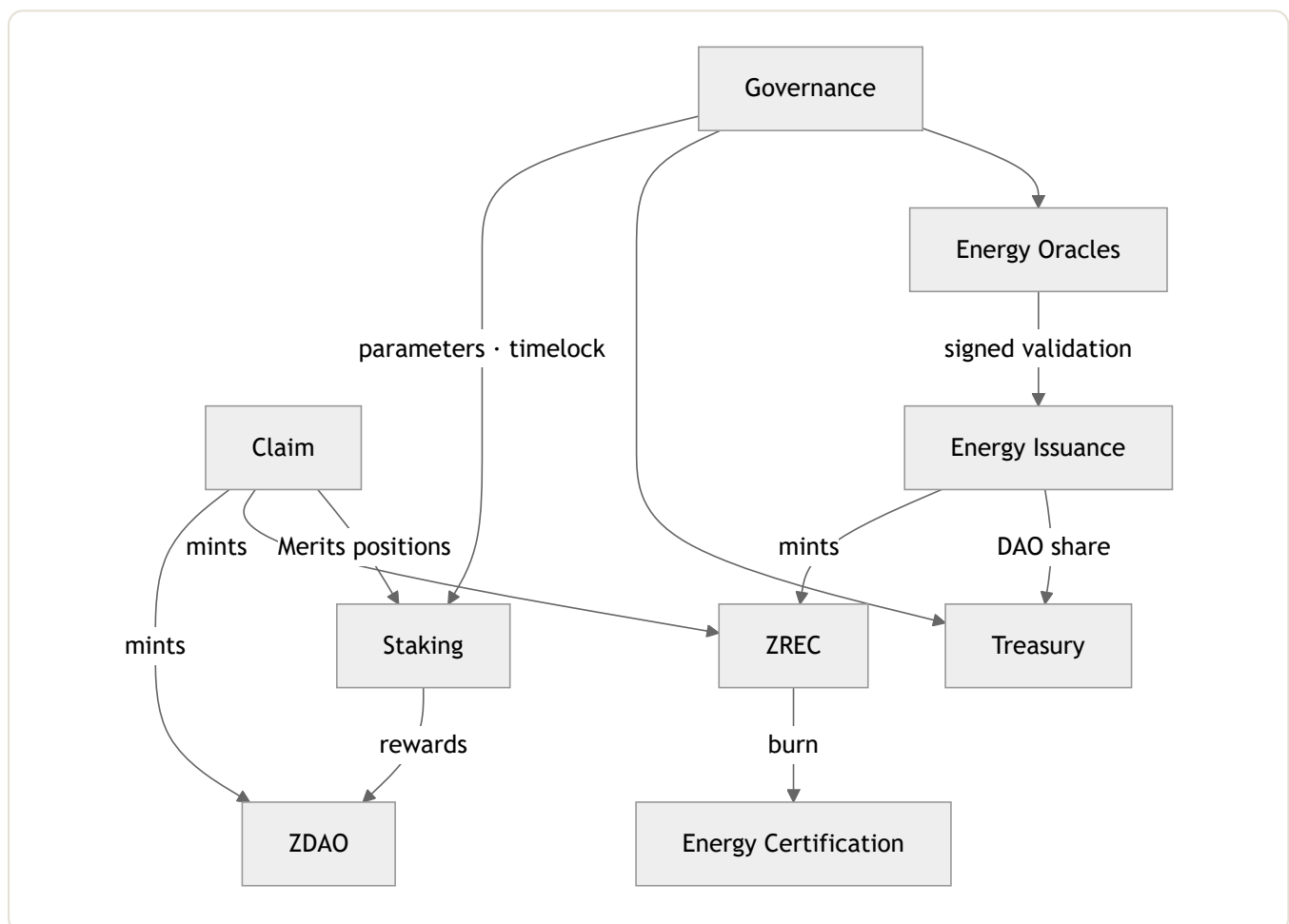
The degree of decentralization of a protocol is not declared: it is demonstrated by the list of capabilities that have been verifiably renounced.

9. CONTRACT ARCHITECTURE

Modular infrastructure on Base, with decoupled responsibilities.

Contract	Main responsibilities
ZERO DAO (ZDAO)	ERC-20, 8 decimals, mintable and burnable. <code>MAX_SUPPLY</code> enforced against <code>totalSupply</code> . Minting restricted to authorized addresses. Burning returns minting capacity
ZERO REC (ZREC)	ERC-20, 3 decimals, single fungible asset. Minting from the Claim (Presale and Merits) and from Energy Issuance. Differentiated Genesis / Energy accounting
Claim	Verification of Merkle proofs against the published root. Resolution of the TOKEN ZERO minting right (ZDAO/ZREC choice) with single consumption. Delivery of ZDAO (Presale) or minting of ZREC (Presale and Merits), with automatic staking for Merits positions
Energy Issuance	Verification of the oracle's validation. Uniqueness registry by batch identifier. Batch status and provisional window. Minting and distribution

Contract	Main responsibilities
Staking	Individualized positions. Reward model linear over time with a rate that halves each year (100% year 1, 50% year 2...). Withdrawal rules differentiated by position type. Cumulative stakingMinted counter capped at the 500M reserve and bounded by the decreasing curve
Energy Certification	Burning of ZREC allocated to offsetting. Issuance of the NFT with its purpose. Non-transferability of Genesis Offset certificates
Governance	Proposals and votes. Parameters within range. Appointment and revocation of the Committee. Execution subject to timelock
Treasury	Multisig custody. Receipt of the energy share of ZREC. Exclusive execution of approved operations
Energy Oracles	Registry of authorized oracles. Signed authorizations with timestamp and expiry. Auditable record of validations



Data acquisition layer. Off-chain infrastructure (Section 4.4). It is not part of the set of contracts and **holds no permission whatsoever over them**. Its replacement or unavailability does not affect the integrity of assets already issued.

The architecture must allow new contracts, services and integrations to be incorporated without replacing the core contracts.

10. SECURITY

10.1 Audit

All contracts must undergo an independent audit before going into production, expressly verifying the invariants of Section 11. Any relevant modification requires an additional audit prior to deployment.

10.2 Vectors considered

Vector	Measure	Section
Repeated Claim after transferring ZDAO	The choice resides in the off-chain allocation, with single consumption	5.3
Duplication of the allocation using several wallets	Terminal state per allocation	5.3
Issuance of rewards above the reserve	The staking reserve keeps its own cumulative counter, which burning does not reduce	2.2
Circumvention of the Merits schedule	The lock lives in the position's withdrawal rules	5.5 and 6.4
Double minting by retrying a batch	Batch identifier in the uniqueness registry	4.6
Reuse of already processed production	Irreversible FINALIZED state	4.6
Double use of a certificate	Registered purpose and non-transferability	3.6
Inflated production declaration	Plausibility checks	4.5
Non-existent or oversized installation	Triple verification of nominal power	4.5
Discretionary authorization of minting	Merkle tree published before the Claim	5.6
Cumulative rounding deviation	Mandatory carry-over of residues	4.6
Accelerated depletion of the staking reserve	Decreasing maximum curve and 500M reserve as a ceiling	6.3

10.3 Treasury and operational security

The DAO Treasury operates under a multisig scheme; no Committee member can dispose of the funds individually; every operation complies with the authorization rules and the timelock; all operations are recorded on-chain.

Operational measures: separation of responsibilities, permission control, permanent record of operations, cryptographic validation of critical actions, protection against replay attacks and duplicate executions, and continuous monitoring.

10.4 Continuous improvement

Governance may incorporate new audits, infrastructure improvements, improvements in permission management, new cryptographic mechanisms and incident response procedures.

The protocol does not require oracles to post a collateral deposit. A deposit does not prove that an energy datum is true: it only introduces an economic penalty if someone cheats. For ZERO Energy, the technical verification chain —installation, meter, measurement, period, validation, uniqueness— described in Section 4 is more decisive. Governance may in the future incorporate economic guarantee or penalty mechanisms for certain validators, without making them a requirement of the protocol now.

11. INVARIANTS AND PARAMETERS

This section gathers in a single place what cannot change and what can change, with its range.

11.1 Invariants

They cannot be modified by any governance mechanism or by contract upgrades. They are implemented so that their verification does not depend on upgradable code.

	Invariant	Section
I-1	The ZDAO totalSupply never exceeds 1,000,000,000 (MAX_SUPPLY of the ERC-20)	2.1
I-2	Burning reduces totalSupply and returns minting capacity under MAX_SUPPLY; it does not replenish the staking reserve	2.2
I-3	500,000,000 ZDAO are reserved exclusively for the Official Staking System (stakingMinted capped at 500M) and only the staking contract can mint them	2.3 and 6.3
I-4	The decreasing issuance curve (250/125/62.5/31.25M) is an absolute annual ceiling	6.3
I-5	A Genesis allocation is consumed only once; the choice between ZDAO and ZREC resides in the off-chain registry, never in the token	5.3
I-6	ZREC is a single fungible asset; restrictions live in the positions, never in the token	3.2
I-7	No unit of energy can support more than one issuance or more than one certificate	4.2
I-8	A batch in FINALIZED state cannot be reused	4.6
I-9	No minting can be authorized on the basis of a non-reproducible criterion	4.5
I-10	The DAO is the supreme authority; the Committee only exercises delegated competences	8.1

	Invariant	Section
I-11	The staking reward accrues linearly over time with a rate that halves each year (100% year 1, 50% year 2...)	6.2

11.2 Configurable parameters

Modifiable by governance within the indicated ranges and subject to the timelock of Section 8.3.

Parameter	Initial value	Range	Section
Energy Issuance split	50% producer	30% – 70%	3.3
Batch closure — time	24 hours	configurable	4.6
Batch closure — energy	1 MWh	configurable	4.6
Provisional window	72 hours	configurable downward	4.6
Burn Execution Threshold	10,000 ZDAO	configurable	2.6
ZERO DEX fee	0.30%	0.05% – 1.00%	7.3
Fee split	40/30/20/10	see table 7.3	7.3
Committee mandate	24 months	renewable	8.1
Staking immediate withdrawal fee	2%	0.50% – 5.00%	6.6
Withdrawal fee split	40/40/20/0	see Section 6.6	6.6
Waiting period and window of the requested withdrawal	set by governance	with safety limits	6.6

12. GENERAL PROVISIONS

12.1 Purpose and interpretation

This document defines the economic, technical and functional architecture of the protocol and constitutes the reference specification for its development and evolution.

During development, technical adaptations may be made provided that they do not alter the invariants of Section 11.1, do not compromise the functional separation between ZDAO and ZREC, do not reduce the security guarantees and are compatible with the decisions approved by governance.

No technical adaptation may affect an invariant. Any adaptation that modifies the described behavior requires the express approval of governance and the corresponding extension of this specification, which will be incorporated as a dated annex.

12.2 Evolution

The incorporation of new functionalities, economic mechanisms, energy integrations, markets or services is carried out through proposals approved by governance. The modular architecture allows such evolution without compromising compatibility with the existing infrastructure.

12.3 Transparency

All protocol activity is carried out in accordance with the principles of transparency, traceability, security, decentralization, sustainability, scalability and auditability. Operations executed through the contracts must be publicly verifiable on Base.

12.4 Vision

ZERO Energy builds a decentralized infrastructure that connects funding, renewable energy production, asset tokenization, economic incentives, governance and environmental certification within a single ecosystem, on the Base network.